

2. What is the expected return of an investor who purchases the discount bond at EUR95.72 and holds it for the remaining four years?

Solution:

1.10 percent

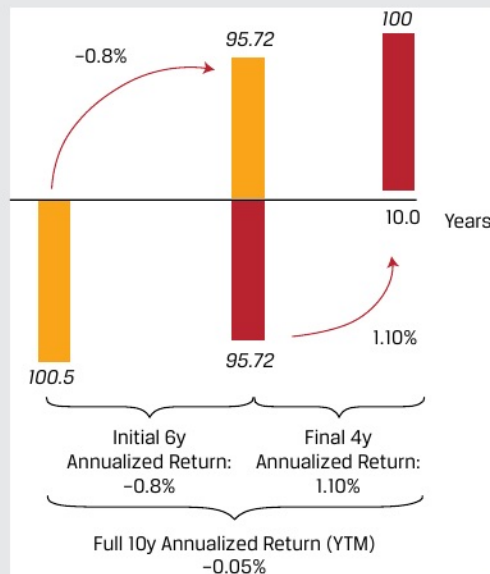
Here the current price of 95.72 in Equation 18 is now the *PV*, and the principal of 100 is *FV*, with $t = 4$:

$$r = 1.10\% = \left(\frac{100}{95.72}\right)^{\frac{1}{4}} - 1.$$

Note that r of 1.10 percent is equal to the YTM for the remaining four years we derived in Example 1. The cumulative returns across the two investors is equal to the initial -0.05 percent YTM, assuming no transaction costs as follows:

$$-0.05\% = [(1 - 0.0081)^6 \times (1 + 0.011)^4]^{\frac{1}{10}} - 1.$$

These relationships are summarized in the following diagram:



Unlike discount bonds, fixed-income instruments that pay periodic interest have cash flows throughout their life until maturity. The uniform discount rate (or internal rate of return) for all promised cash flows is the YTM, a single implied market discount rate for all cash flows regardless of timing as shown for a coupon bond in Exhibit 8.